



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)  
UNIT TEST  
21-06-2020

## JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

Time : 3 Hours

Maximum Marks : 180

### READ THE INSTRUCTIONS CAREFULLY

#### GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

#### DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

**SOME USEFUL CONSTANTS**

**Atomic No.** : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses** : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- |                                    |                                                              |
|------------------------------------|--------------------------------------------------------------|
| • Boltzmann constant               | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$                   |
| • Coulomb's law constant           | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$                   |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$  |
| • Speed of light in vacuum         | $c = 3 \times 10^8 \text{ ms}^{-1}$                          |
| • Stefan-Boltzmann constant        | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$                        |
| • Permeability of vacuum           | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$                |
| • Permittivity of vacuum           | $\epsilon_0 = \frac{1}{\mu_0 c^2}$                           |
| • Planck constant                  | $h = 6.63 \times 10^{-34} \text{ J-s}$                       |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

## PART-1 : PHYSICS

### SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.  
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).  
*Negative Marks* : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A charged particle moves in a viscous environment with constant velocity in mutually perpendicular uniform electric and magnetic fields equal to  $E$  and  $B$  respectively. Find the angle  $\theta$  between velocity and the direction  $B$  if force of resistance is proportional to the speed. It is known that if only electric field was present, the steady state speed of the particle was  $V_0$ . Neglect Gravity. Also find the angle  $\phi$  between electric field and the velocity of the particle.

(A) velocity has no component in the direction of magnetic field.

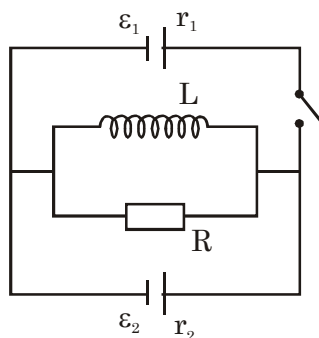
(B)  $\theta = \tan^{-1} \left( \frac{E}{BV_0} \right)$

(C)  $\phi = \tan^{-1} \left( \frac{BV_0}{E} \right)$

(D)  $\phi = \tan^{-1} \left( \frac{2BV_0}{E} \right)$

Space for Rough Work

2. In the given figure:  $\varepsilon_1 = \varepsilon_2 = 10 \text{ V}$ ;  $r_1 = 5 \text{ Ohm}$ ;  $r_2 = 20 \text{ Ohm}$ ;  $R = 4 \text{ Ohm}$ . The key is initially open. The key is closed at  $t = 0$ .



- (A) The current through the inductor before  $t = 0$  is  $0.5 \text{ A}$   
 (B) The current through the resistor  $R$  immediately after closing the switch is  $1 \text{ A}$   
 (C) The current through the battery  $\varepsilon_2$  after a long time is  $2 \text{ A}$   
 (D) The current through the inductor after a long time is  $2.5 \text{ A}$
3. A wire of length  $314 \text{ cm}$  carries a current of  $3 \text{ A}$  when connected to a battery. It can be made into (i) single loop shaped like a circle (ii) single loop shaped like a square, (iii) single loop shaped like an equilateral triangle or (iv) flat compact circular coil of  $10$  turns. Magnetic moments in these cases are compared. (Take :  $\pi = 3.14$ )

(A)  $M_{\max} = \frac{3\pi}{4} \text{ Am}^2$

(B)  $M_{\min} = \frac{3\pi}{40} \text{ Am}^2$

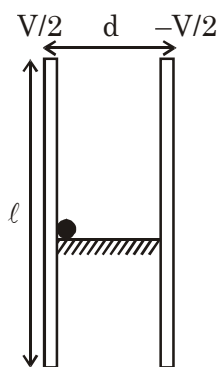
(C)  $M_{\text{square}} = M_{\text{triangle}} \times \frac{\sqrt{3}}{4}$

(D)  $M_{\text{flat coil}}$  is minimum

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Space for Rough Work

4. Two large square metallic plates ( $\ell \times \ell$ ) are separated by a distance  $d$ . The potential of plates remain constant at  $\frac{V}{2}$  &  $-\frac{V}{2}$ . The plates are connected by a smooth horizontal insulating platform. A very small metallic ball of mass  $m$  and radius  $r$  can move on the platform. Assume that when it touches a plate, it instantaneously acquires the same potential as that of the plate. The collision are completely inelastic.  $\ell \gg d \gg r$

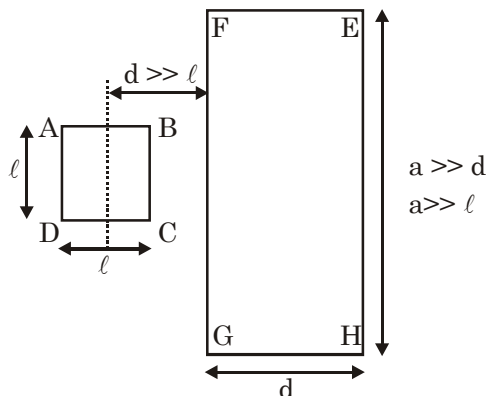


- (A) The effective charge that flows from left plate to right plate in a complete round trip is  $4\pi\epsilon_0 Vr$ .
- (B) The average current over a large period of time is proportional to  $V^{3/2}$ .
- (C) The "resistance" to current flow  $\left(\frac{V}{i}\right)$  is inversely proportional to  $V$ .
- (D) The average power loss is proportional to  $V^2$ .

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**Space for Rough Work**

5. A square conducting loop ABCD and a rectangular conducting loop EFGH are placed on the same plane as shown in the figure. The loop ABCD is placed symmetrically with respect to EFGH such that its central line is at a distance  $d$  from the loop EFGH. Side length ' $\ell$ ' of small loop ABCD is much smaller than the length ' $a$ ' and width ' $d$ ' of the bigger loop EFGH. A source is applied on loop ABCD, such that it sets up a current  $I = kt$  in it. Here  $k$  is a positive constant and  $t$  is time. Then choose the **CORRECT** option(s)

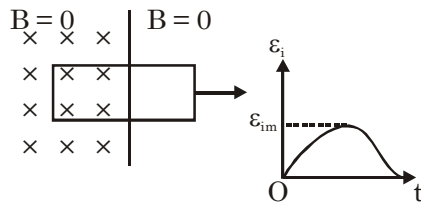


- (A) If the current in the loop ABCD is in clockwise sense then the induced current in the loop EFGH due to this current is in anticlockwise sense.
- (B) If the current in the loop ABCD is in clockwise sense then the induced current in the loop EFGH due to this current is in clockwise sense.
- (C) emf induced in the loop EFGH due to current in the loop ABCD is  $\frac{\mu_0 \ell^2 k}{4\pi d}$ .
- (D) emf induced in the loop EFGH due to current in the loop ABCD is  $\frac{3\mu_0 \ell^2 k}{4\pi d}$ .

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Space for Rough Work

6. Plane rectangular loop is placed in a magnetic field. The loop is pulled out of the magnetic field at a variable velocity. Assume the  $\vec{B}$  is uniform and constant. The emf induced in the loop due to this field is  $\varepsilon_i$  whose maximum value is  $\varepsilon_{im}$ .  $\varepsilon_i$  is plotted against time as shown in graph. Which of the following are is **CORRECT** statement(s) :-



- (A)  $\varepsilon_{im}$  is independent of rate of removal of coil from the field.  
 (B) The total charge that passes through any point of the loop in the process of complete removal of the loop does not depend on velocity of removal.  
 (C) The total area under the curve ( $\varepsilon_i$  vs  $t$ ) is independent of rate of removal of coil from the field.  
 (D) The area under the curve is dependent on the rate of removal of the coil.

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Space for Rough Work

**SECTION-I(ii) : (Maximum Marks : 12)**

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +3 If only the bubble corresponding to the correct answer is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases

**Paragraph for Questions 7 and 8**

A solid sphere of radius  $R$  carries a uniform charge density,  $+\rho_0$ , between  $r = 0$  and  $r = R_s$  and an equal charge density of opposite sign,  $-\rho_0$ , between  $r = R_s$  and  $r = R$ . The sphere rotates about its diameter with angular velocity  $\omega$ .

7. What is the electric field at the surface of sphere (at a distance  $R$  from centre).

(A)  $\frac{\rho_0}{3\epsilon_0} \left( \frac{R_s^3}{R^2} - R \right)$     (B)  $\frac{\rho_0}{3\epsilon_0} (R_s + R)$     (C)  $\frac{\rho_0}{3\epsilon_0} \left( \frac{2R_s^3}{R^2} - R \right)$     (D)  $\frac{\rho_0}{6\epsilon_0} \left( \frac{R_s^3}{R^2} - R \right)$

8. What is value of  $R_s$  such that magnetic moment of the sphere is zero ?

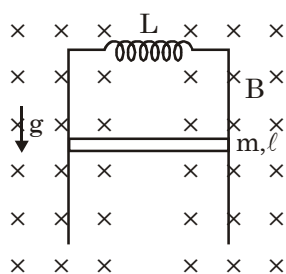
(A)  $\frac{R}{2}$     (B)  $\frac{R}{\sqrt[5]{2}}$     (C)  $\frac{R}{\sqrt[3]{2}}$     (D)  $\frac{R}{\sqrt[4]{2}}$

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**Space for Rough Work**

**Paragraph for Questions 9 and 10**

A coil of inductance  $L$  connects the upper end of two vertical copper bars separated by a distance  $\ell$ . A horizontal conducting connector of mass  $m$  starts falling with zero initial velocity along the bars without losing contact with them. The whole system is located in uniform, magnetic field with induction  $B$  perpendicular to the plane of the bars. There is no resistance in the circuit and no friction.



9. What is maximum velocity of connector ?

- (A)  $\frac{2g}{\ell B} \sqrt{mL}$       (B)  $\frac{g}{\ell B} \sqrt{mL}$       (C)  $\frac{g}{3\ell B} \sqrt{mL}$       (D)  $\frac{4g}{\ell B} \sqrt{mL}$

10. What is maximum current in inductor ?

- (A)  $\frac{mg}{B\ell}$       (B)  $\frac{2mg}{B\ell}$       (C)  $\frac{4mg}{B\ell}$       (D)  $\frac{3mg}{B\ell}$

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**Space for Rough Work**

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

| + |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ● | ● | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ |
| ① | ① | ① | ① | ① | ① | ① | ① | ① | ① |
| ② | ② | ② | ② | ● | ② | ② | ② | ② | ② |
| ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ |
| ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ |
| ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ |
| ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ |
| ⑦ | ⑦ | ● | ● | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ |
| ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ |
| ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ |

| - |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ● | ● | ● | ○ | ○ | ○ | ○ | ○ | ○ | ○ |
| ① | ① | ① | ① | ① | ① | ① | ① | ① | ① |
| ② | ② | ② | ② | ● | ② | ② | ② | ② | ② |
| ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ |
| ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ |
| ⑤ | ⑤ | ⑤ | ● | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ |
| ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ |
| ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ |
| ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ |
| ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ |

- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +3 If **ONLY** the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

1. Fuse wire will melt if the voltage across it is equal to 10 V. At what voltage will the fuse melt if its length is doubled keeping the radius and material same? Assume Newton's law of cooling, and assume that the heat is lost only by radiation.
2. A long straight wire lies on y-axis and carries a current of 8 A in -y direction. In addition to the magnetic field produced by the wire, a uniform magnetic field of  $1.4 \times 10^{-6} \text{ T} \hat{i}$  is also present. What is the magnitude of net magnetic field at (1, 1, 1). Express your answer in  $\mu\text{T}$ .

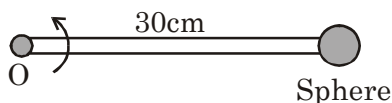
Space for Rough Work

3. Charge distribution with spherical symmetry of radius  $R$  has charge density  $\rho = kr^{-n}$ , where  $k$  and  $n$  are positive constants and  $r < R$ . Calculate the electrostatic potential energy of this charge distribution. For what integral value of  $n$  is the energy infinite ?
4. A cylindrical region has magnetic field along its axis. The magnetic field is cylindrically symmetric and its value is given by  $B = B_0 t(1 - r/R)$ . Where  $R = 0.8$  m is the radius of the cylindrical region,  $B_0 = 0.5$  T/sec and  $t$  is time. At what distance (in cm) from the axis of the cylinder is the induced electric field maximum ?

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**Space for Rough Work**

5. Two long wires are placed on a pair of fixed parallel rails perpendicular to the rails. The spacing  $d$  between the rails is large compared with  $x$ , the distance between the wires. Both wires and rails are made of a material of resistance  $\rho$  per unit length. A magnetic flux density  $B$  is applied perpendicular to the rectangle so formed, and one wire is moved along the rails with a uniform speed  $v$  while the other is held stationary. It is seen that the force required to hold the stationary wire vanishes at a particular value of  $x$ . Find  $x$  (in m).  
[Take :  $\rho = 10^{-6} \Omega/\text{m}$ ,  $v = 4 \text{ m/s}$ ,  $B = 1 \text{ T}$ ]
6. A sphere of radius 1 cm is charged to a voltage of 900V. The sphere is mounted to a 30 cm long insulating handle and is rotated about O, the number of revolutions is 1800/ minute. What is the magnetic induction (in  $10^{-14} \text{ T}$ ) which can be observed at the position 'O' of the axis of the rotation?




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Space for Rough Work

7. An oscillation circuit consists of a coil of inductance  $L$  with negligible resistance and flat plate capacitor of capacity  $C$ , the distance between the plates of which is equal to  $d$ . The capacitor plates are not charged, and there is no current in the circuit. For a short time  $\Delta t \ll T$  in the region of the capacitor, we created an electric field  $E$  directed perpendicularly to the plates (By connecting it to a cell). The coil remained out electric field. What will be the maximum current (in A) in the circuit thereafter ? Take  $L = 1 \text{ mH}$ ,  $C = 1000 \mu\text{F}$ ,  $E = 1000 \text{ N/C}$ ,  $d = 1 \text{ mm}$ ,  $\Delta T = 10^{-5} \text{ sec}$
8. A particle of mass  $1 \times 10^{-26} \text{ kg}$  and charge  $+1.6 \times 10^{-19} \text{ C}$  travelling with a velocity of  $1.28 \times 10^6 \text{ m/s}$  in the  $+x$  direction enters a region in which a uniform electric field  $E$  and a uniform magnetic field of induction  $B$  are present such that  $E_x = E_y = 0$ ,  $E_z = -102.4 \text{ kV/m}$  and  $B_x = B_z = 0$ ,  $B_y = 8 \times 10^{-2} \text{ Wb m}^{-2}$ . The particle enters this region at the origin at time  $t = 0$ . If X-coordinate of the particle is  $x$  (in m) at  $t = 5 \times 10^{-6} \text{ s}$ . Then find value of  $x$ .

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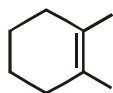
**Space for Rough Work**

## PART-2 : CHEMISTRY

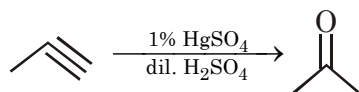
### SECTION-I(i) : (Maximum Marks: 24)

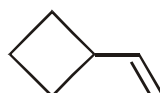
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1. Which of the following statement(s) is/are correct?

(A) Total number of monobrominated products obtained on reaction of  with NBS are 7.

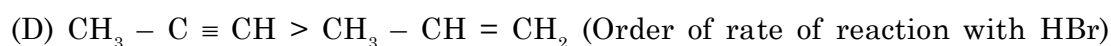
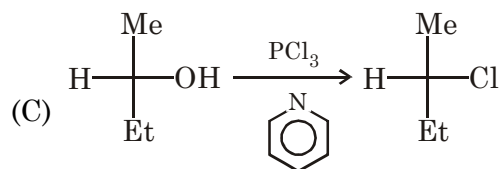
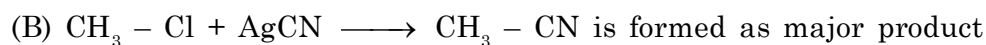
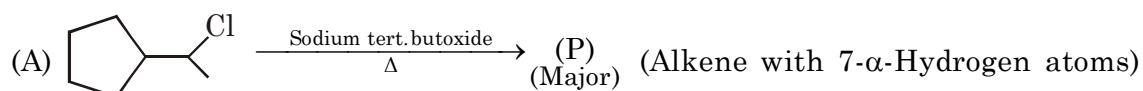
(B)  $\text{H}-\overset{\text{O}}{\underset{\text{O}}{\parallel}}{\text{C}}-\text{H} > \text{CH}_3-\text{C}(\text{CH}_3)=\text{O} > \text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} > \text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{OR}$  (Order of reactivity with  $\text{CH}_3\text{MgCl}$ )

(C) 

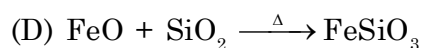
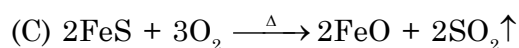
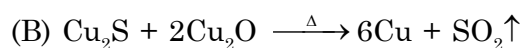
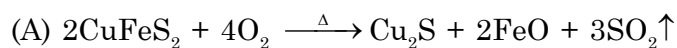
(D)  (P), product (P) will give instant turbidity with Lucas reagent.

Space for Rough Work

2. Incorrect statement(s) in the following sequence is/are



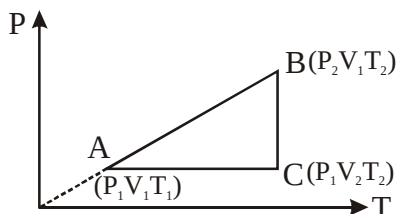
3. Which of the following reaction(s) does/do occur in any step during the extraction of pure copper from copper pyrite?




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Space for Rough Work

4. Which of the following complex show optical isomerism and spin paired (low spin) complex?  
 (A)  $[\text{Co}(\text{en})_3]\text{Cl}_3$  (B)  $\text{Na}_3[\text{Cr}(\text{ox})_3]$  (C)  $[\text{Pt}(\text{NH}_3)(\text{H}_2\text{O})\text{Cl}_2]$  (D)  $\text{Na}_3[\text{Co}(\text{gly})_3]$
5. A reversible cyclic process for an ideal gas is shown below. Here P, V and T are pressure, volume and temperature, respectively. The thermodynamic parameters q, w, H and U are heat, work, enthalpy and internal energy, respectively ,

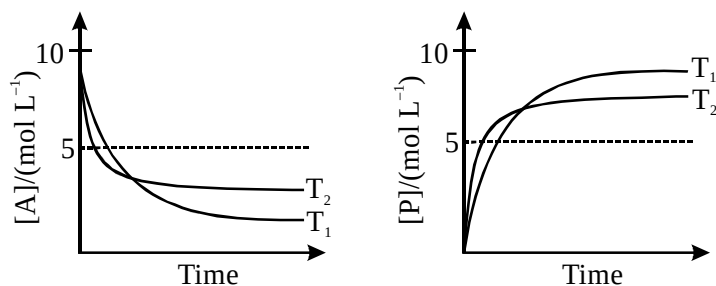


The correct option(s) is (are) -

- (A)  $q_{AB} = \Delta U_{AC}$  and  $W_{CA} + W_{AB} = -W_{BC}$   
 (B)  $\Delta U_{BC} = \Delta H_{BC}$  and  $\Delta U_{CA} + \Delta U_{AB} = -\Delta U_{BC}$   
 (C)  $q_{BC} = W_{AB}$  and  $q_{CA} + W_{CA} + q_{AB} + w_{AB} = q_{CB} + w_{CB}$   
 (D)  $q_{ABCA} = w_{ACBA}$  and  $\Delta H_{ABCA} = \Delta U_{ACBA}$

Space for Rough Work

6. For a reaction,  $A \rightleftharpoons P$ , the plots of  $[A]$  and  $[P]$  with time at temperatures  $T_2$  and  $T_1$  are given below.



If  $T_2 < T_1$ , the correct statement(s) is (are)

(Assume  $\Delta H^\theta$  and  $\Delta S^\theta$  are independent of temperature and ratio of  $\ln K$  at  $T_1$  to  $\ln K$  at  $T_2$  is greater than  $\frac{T_2}{T_1}$ . Here H, S, G and K are enthalpy, entropy, Gibbs energy and equilibrium constant, respectively.)

(A)  $\Delta H^\theta < 0$ ,  $\Delta S^\theta < 0$

(B)  $\Delta G^\theta < 0$ ,  $\Delta H^\theta > 0$

(C)  $\Delta G^\theta < 0$ ,  $\Delta S^\theta < 0$

(D)  $\Delta G^\theta < 0$ ,  $\Delta S^\theta > 0$

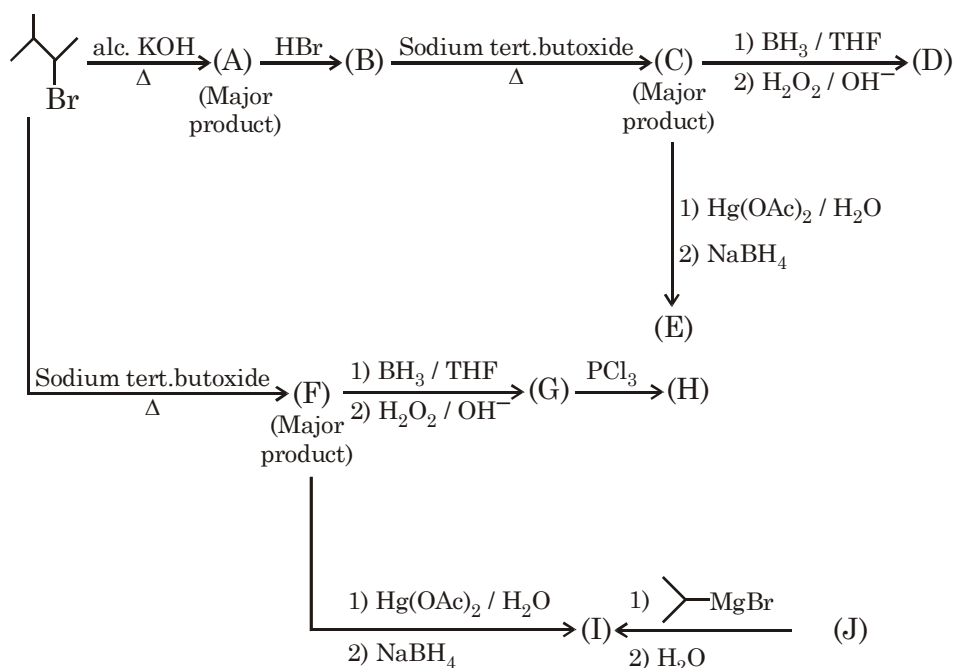
---

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +3 If only the bubble corresponding to the correct answer is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases

Paragraph for Questions 7 and 8

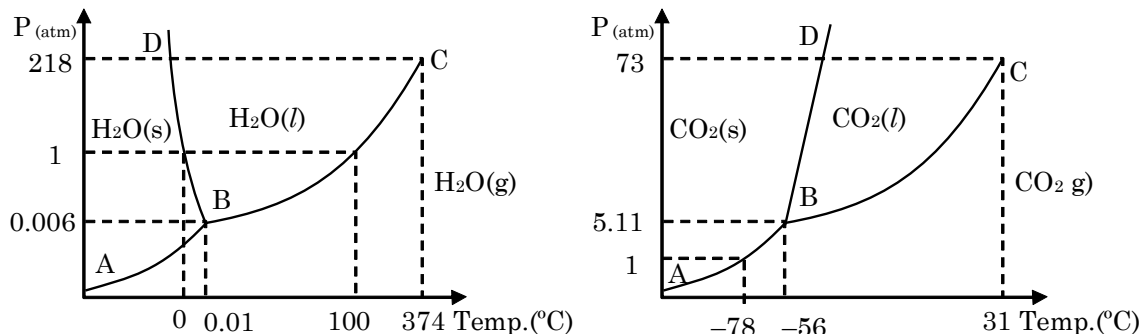


7. Correct statement for above reaction sequence?
- (A) (B) to (C) is substitution reaction  
 (B) Compound (E) shows instant turbidity with Lucas reagent  
 (C) (J) is acetone  
 (D) (A) to (B) is free radical addition.
8. Incorrect statement for above reaction sequence?
- (A) Compound (D) is a 1°-alcohol  
 (B) Compound (I) is a 1°-alcohol  
 (C) Compound (E) is a 3°-alcohol  
 (D) Compound (H) is a 1°-alkyl halide

Space for Rough Work

Paragraph for Questions 9 and 10

Phase diagram summarizes the effect of temperature and pressure on a substance in a closed container. Every point in this diagram represents a possible combination of temperature and pressure for the system. The diagram is divided into three areas, which represents the solid, liquid and gaseous states of the substance.



The best way to remember which area corresponds to each of these states is to remember the conditions of temperature and pressure that are most likely to be associated with a solid, a liquid and a gas. Low temperature and high pressures favor the formation of a solid. Gases on the other hand are most likely to be found at high temperatures and low pressure. Liquids lie between these extremes. AB represents solid and gas equilibrium, BD represents solid and liquid equilibrium, BC represents liquid and gas equilibrium.

9. Select the **CORRECT** option based on above phase diagram.

- (A)  $(\Delta_f G^0)_{\text{CO}_2(\text{s})} > (\Delta_f G^0)_{\text{CO}_2(\text{g})}$  at 1 atm & 298 K
- (B)  $(\Delta_f H)_{\text{CO}_2(\text{s})} = (\Delta_f H)_{\text{CO}_2(\text{g})}$  at 5.11 atm &  $-56^\circ\text{C}$
- (C)  $(\Delta_f S^0)_{\text{CO}_2(\text{s})} = (\Delta_f S^0)_{\text{CO}_2(\text{g})}$  at 1 atm &  $-78^\circ\text{C}$
- (D)  $\Delta_r S$  for  $\text{CO}_2(\text{l}) \rightarrow \text{CO}_2(\text{g})$  increases on increasing pressure at 298K

10. Select the **INCORRECT** option based on above phase diagram.

- (A)  $\Delta_r G$  for  $\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{H}_2\text{O}(\text{l})$ ,  $\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{H}_2\text{O}(\text{g})$  are equal at 0.006 atm and  $0.01^\circ\text{C}$ .
- (B)  $\Delta_r G$  at  $0.01^\circ\text{C}$  for  $\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{H}_2\text{O}(\text{l})$  is less at 1 atm compared to 0.006 atm
- (C)  $\Delta S_{\text{sublimation, m}} = \Delta S_{\text{fusion, m}} + \Delta S_{\text{vaporisation, m}}$  at triple point conditions.
- (D)  $\Delta_f G^0 \text{H}_2\text{O}(\text{l}) < \Delta_f G^0 \text{H}_2\text{O}(\text{g})$  at 1 atm,  $374^\circ\text{C}$ .

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

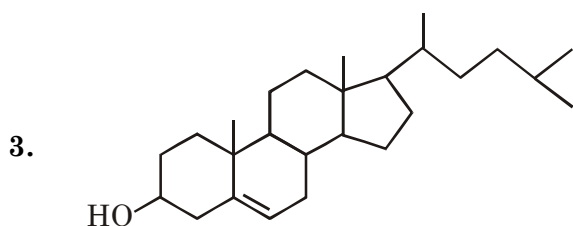
**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

| + |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

| - |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +3 If **ONLY** the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

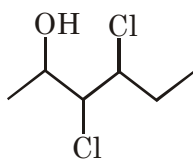
- Two moles of an ideal diatomic gas is subjected to an isothermal increase in pressure from 1 atm to 10 atm at 300K. The change in Gibb's free energy of the system (in kJ) is. (R = 8.3 J/K-mol)
- 530 gm, **1 molal** aq. urea solution is mixed with 402 gm, **2 molal** aq. NaBr solution if the resulting solution has sp gravity 466/375 at 300K, calculate osmotic pressure of resulting solution ( in atm).  
**[Given : R = 0.08 l-atm/mol-K, Br = 80)**



Identify total number of chiral centre in above structure?

Space for Rough Work

4. Identify total number of stereo isomers for following compound?

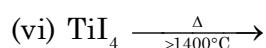
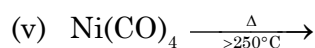
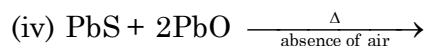
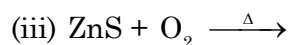


5. What is the average oxidation state of sulphur in Chalcopyrite?
6. Among the following ores  
Galena, Calamine, Bauxite, Argentite, Carnelite, Tin stone  
If 'X' is the number of ore(s) for which electrometallurgical operation is used for reduction of metal ion and 'Y' is the number of ore(s) for which pyrometallurgical operation is used for reduction of metal ion then find the value of  $\left(\frac{X}{Y}\right)$

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**Space for Rough Work**

7. During the metallurgy of various metals from their respective ores, some reactions are involved which are given as :



If 'X' is the number of given reaction(s) in which redox reaction takes place and 'Y' is the number of given reaction(s) in which thermal refining of impure metal is involved then find the value

of  $\left(\frac{X+1}{Y}\right)$ ?

8. If 'A' and 'B' are respectively optically active and optically inactive isomers in  $[\text{Pt}(\text{NH}_3)_2(\text{H}_2\text{O})_2\text{ClBr}]\text{I}$  and 'C' and 'D' are respectively geometrical isomers and optical isomers in  $[\text{Co}(\text{gly})_3]$ .

Then find the value of  $\left(\frac{A+B}{C+D}\right)$  ?

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Space for Rough Work

**PART-3 : MATHEMATICS****SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.  
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).  
*Negative Marks* : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1.  $\int \frac{dx}{\sqrt{2} + \sin x + \cos x}$  is equal to -

(A)  $\frac{1}{\sqrt{2}} \left( \sec \left( x + \frac{\pi}{4} \right) - \tan \left( x + \frac{\pi}{4} \right) \right) + C$

(B)  $\frac{1}{\sqrt{2}} \left( \sec \left( x - \frac{\pi}{4} \right) - \tan \left( x - \frac{\pi}{4} \right) \right) + C$

(C)  $\frac{1}{\sqrt{2}} \left( \tan \left( x + \frac{\pi}{4} \right) - \sec \left( x + \frac{\pi}{4} \right) \right) + C$

(D)  $\frac{1}{\sqrt{2}} \left( \tan \left( x - \frac{\pi}{4} \right) - \sec \left( x - \frac{\pi}{4} \right) \right) + C$

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**Space for Rough Work**

2. Let S be the area bounded by  $y = e^{|\cos 4x|}$ ,  $x = 0$ ,  $y = 0$  and  $x = \pi$ . Then

(A)  $S = 2 \int_0^{\pi/2} e^{\sin t} dt$       (B)  $S > \frac{\pi}{2}$       (C)  $S < 2 \left( e^{\frac{\pi}{2}} - 1 \right)$       (D)  $S \leq 1$

3. If  $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ , then which of the following is(are) INCORRECT

(where  $A^{-1}$  is inverse of matrix A, I is an unit matrix of order 3, adj A is adjoint of matrix A,  $|A|$  denotes determinant value of matrix A.)

(A)  $\frac{1}{81} |\text{adj}(2A) \cdot \text{adj}(3A^{-1}) \cdot A^{-1}|$  is equal to 288

(B)  $(A^{-1})^2 = \frac{1}{2}(A - 5I + 7A^{-1})$

(C)  $|A^5 - 2A^4|$  is equal to 16

(D)  $A^{-1} = \frac{1}{2}(A^2 - 5A + 6I)$

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Space for Rough Work

4. The vectors  $\vec{p}, \vec{q}, \vec{r}$  are of the same length and pairwise form equal angles. If  $\vec{p} = \hat{i} + \hat{j}$  and  $\vec{q} = \hat{j} + \hat{k}$ , then the vector  $\vec{r}$  can be-
- (A)  $\left(-\frac{4}{3}\hat{i} + \frac{1}{3}\hat{j} - \frac{4}{3}\hat{k}\right)$  (B)  $(\hat{i} + \hat{k})$  (C)  $\left(-\frac{1}{3}\hat{i} - \frac{4}{3}\hat{j} - \frac{1}{3}\hat{k}\right)$  (D)  $\left(-\frac{1}{3}\hat{i} + \frac{4}{3}\hat{j} - \frac{1}{3}\hat{k}\right)$
5. If the line  $x - 1 = 0$  divides the area bounded by the curves  $2x + 1 = \sqrt{4y + 1}$ ,  $y = x$  and  $y = 2$  in 2 regions of area  $A_1$  and  $A_2$  ( $A_1 < A_2$ ), then
- (A)  $A_1 = \frac{1}{4}$  (B)  $A_1 = \frac{1}{3}$  (C)  $A_2 = \frac{1}{2}$  (D)  $A_2 = \frac{1}{4}$
6. If  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuous function satisfying the equation  $2xf(x) + (1 + x^2)f'(x) = 1$  and  $f(0) = 0$ , then-
- (A)  $f(x)$  is an odd function
- (B) Number of integers in the range of  $f(x)$  is 1.
- (C) Number of solutions of  $f(x) = \frac{1}{4}$  are 2
- (D) Number of solutions of  $f(f(x)) = \frac{1}{2}$  are 2.

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Space for Rough Work

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SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
  - Based on each paragraph, there are **TWO** questions.
  - Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
  - For each question, darken the bubble corresponding to the correct option in the ORS.
  - For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +3 If only the bubble corresponding to the correct answer is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases
- 

**Paragraph for Questions 7 and 8**

Let A be set of all  $3 \times 3$  symmetric matrices all of whose entries are either a or b or c, two of these entries are a, four of them are b and three of them are c.

7. The number of matrices A, (if a, b, c are distinct real numbers), is -  
(A) 27 (B) 12 (C) 30 (D) 18
8. If  $a = c = 1$  &  $b = 0$ , then number of matrices A for which the system of linear equations  $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ , has an unique solution -  
(A) 4 (B) 6 (C) 8 (D) 12
- 

**Space for Rough Work**

**Paragraph for Questions 9 and 10**

Let  $g(x) = \frac{x(x-1)}{(x^2+1)} e^{-x}$  and function  $y = f(x)$  satisfy the differential equation  $dy - ydx = e^x g'(x) dx$

also  $f(0) = 0$

9.  $f'(1)$  is equal to-

(A)  $\frac{1}{4}$

(B)  $\frac{1}{2}$

(C)  $\frac{3}{4}$

(D) 1

10. Area bounded by  $f(x)$  and x-axis in IV quadrant is-

(A)  $\ln\left(\frac{e}{2}\right) - \frac{\pi}{4}$

(B)  $\ln\left(\frac{e}{2}\right) - \frac{\pi}{2}$

(C)  $\frac{\pi}{4} - \ln\left(\frac{e}{\sqrt{2}}\right)$

(D)  $1 - \frac{1}{2} \ln 2 - \frac{\pi}{4}$

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**Space for Rough Work**

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

| + |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ● | ● | ● | ● | ● | ● | ● | ● | ● | ● |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

| - |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ● | ● | ● | ● | ● | ● | ● | ● | ● | ● |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +3 If **ONLY** the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

- The value of the integral  $I = \int_1^3 \left( \tan^{-1} \left( \frac{x^2 - 5x + 6}{x^3 - 6x^2 + 12x - 7} \right) + \tan^{-1} \left( \frac{1}{x^2 - 4x + 4} \right) \right) dx$  is
- Let  $I_n = \int \tan^n x dx$ , then  $\lim_{x \rightarrow 0} \frac{(I_{10} - I_6)}{x^7}$ , if it exists, is equal to
- Let ABCD is a parallelogram where E divides AB internally in ratio 3 : 1 and F divides AD internally in ratio 1 : 3. If P is intersection point of ED and CF, then the ratio  $\frac{PE}{PD}$  is equal to

Space for Rough Work

4. If the area bounded by  $y = f(x)$ ,  $x = \frac{1}{2}$ ,  $x = \frac{\sqrt{3}}{2}$  and x-axis is A sq. units where  $f(x) = x + \frac{2}{3}x^3 + \frac{24}{35}x^5 + \frac{246}{357}x^7 + \dots, |x| < 1$ ; then the value of  $[4A]$  is (where  $[.]$  is G.I.F.....)
5. If  $y = f(x)$  ( $y > 0$ ) satisfies the differential equation  $x \frac{dy}{dx} - y = \frac{x^4}{y}$  such that  $y(1) = 2$ , then the value of  $f(-1)$  is
6. Let  $f(x) = x^3 - 3x^2 + 3x + 1$  and  $g$  be inverse of it, then area bounded by the curve  $y = g(x)$  with x axis between  $x = 1$  and  $x = 2$ , is (in square units)

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**Space for Rough Work**

7. Let A is symmetric and B is a skew symmetric matrix, such that  $A - B = \begin{bmatrix} 2 & 5 \\ 7 & 6 \end{bmatrix}$ , then  $\det(A)$  is equal to

8. In a triangle ABC,  $\angle CAB = 90^\circ$ ,  $|\overrightarrow{AB}| = 1$ ,  $|\overrightarrow{AC}| = \sqrt{3}$  and a point P inside the  $\triangle ABC$  satisfies

$$\frac{\overrightarrow{PA}}{|\overrightarrow{PA}|} + \frac{\overrightarrow{PB}}{|\overrightarrow{PB}|} + \frac{\overrightarrow{PC}}{|\overrightarrow{PC}|} = \vec{0}, \text{ then } \frac{|\overrightarrow{PC}| - |\overrightarrow{PB}|}{|\overrightarrow{PA}|} \text{ is equal to}$$

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**Space for Rough Work**

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Space for Rough Work

**QUESTION PAPER FORMAT AND MARKING SCHEME :**

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

| Section | Que. Type                                         | No. of Que. | Category-wise Marks for Each Question                                                   |                                                                                                          |                                         |                          | Maximum Marks of the section |
|---------|---------------------------------------------------|-------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------|------------------------------|
|         |                                                   |             | Full Marks                                                                              | Partial Marks                                                                                            | Zero Marks                              | Negative Marks           |                              |
| I(i)    | One or more correct option(s)                     | 6           | +4<br>If only the bubble(s) corresponding to all the correct option(s) is(are) darkened | +1<br>For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened | 0<br>If none of the bubbles is darkened | -2<br>In all other cases | 24                           |
| I(ii)   | Paragraph Based (Single correct option)           | 4           | +3<br>If only the bubble corresponding to the correct option is darkened                | —                                                                                                        | 0<br>If none of the bubbles is darkened | -1<br>In all other cases | 12                           |
| II      | Numerical Value Type (Up to second decimal place) | 8           | +3<br>If only the bubble corresponding to correct answer is darkened                    | —                                                                                                        | 0<br>In all other cases                 | —                        | 24                           |

NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions and shall abide by them.

 \_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

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Signature of the Invigilator